

Second
Free Module

First
Free Module

Target
Ring

$$0 \longrightarrow R \xrightarrow{\Psi = \begin{bmatrix} -y \\ x \end{bmatrix}} R^2 \xrightarrow{\Phi = \begin{bmatrix} x & y \end{bmatrix}} R$$

$$\begin{array}{ccc} c & \xrightarrow{\text{multiply by } \Psi} & \begin{bmatrix} -cy \\ cx \end{bmatrix} \xrightarrow{\text{multiply by } \Phi} 0 \end{array}$$

The Homological Core: $\Phi\Psi = 0$

$$\begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} -y \\ x \end{bmatrix} = x(-y) + y(x) = -xy + xy = \mathbf{0}$$

The image of Ψ is exactly the kernel of Φ .
The matrix sequence is perfectly resolved.